

Solving with the Quadratic Formula

Step-by-step worked example: $x^2 - x - 6 = 0$

- 1 Set the equation equal to 0**

$$x^2 - x - 6 = 0$$



- 2 Identify a , b , and c**

$$a = 1 \quad b = -1 \quad c = -6$$



- 3 Write the quadratic formula**

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$



- 4 Substitute a , b , c — keep the parentheses!**

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-6)}}{2(1)}$$



- 5 Simplify under the radical**

$$x = \frac{1 \pm \sqrt{1 + 24}}{2} = \frac{1 \pm \sqrt{25}}{2}$$



- 6 Take the square root (look for a perfect square)**

$$x = \frac{1 \pm 5}{2}$$



- 7 Split into two equations**

$$x = \frac{1+5}{2} \quad x = \frac{1-5}{2}$$



- 8 Simplify each**

$$x = 3 \quad x = -2$$

✓ Check by Factoring

$$(x - 3)(x + 2) = 0$$

$$x - 3 = 0 \Rightarrow x = 3$$

$$x + 2 = 0 \Rightarrow x = -2$$

Same answers — it checks!

★ Remember

- Set it = 0 before you start.
- Keep parentheses when you substitute — watch negatives.
- Look for a perfect square under the radical.
- Not a perfect square? The answer is irrational — leave it in exact (radical) form.

SOLUTIONS

$$x = 3 \quad \text{or} \quad x = -2$$